

The W-3 is finished — and the checker that reads the law had a hole in it

Slice books-55 · for Michael Lyman · Greenway Marijuana

The short version

The W-3 is taught, box by box, all thirty-one of them. That was the job you asked for and it is done.

But the more valuable thing in this slice is something I found by accident while proving the W-3 lessons quoted the law correctly. **The tool that checks every legal quotation in this system against the actual government text had a hole in it, and three quotes had been sitting unverified behind a green checkmark for several slices.** The cause was two characters. I fixed it, then built a permanent guard so that particular kind of blindness cannot happen a fifth time.

I say fifth because it had already happened four times. More on that below, because it is the part I think you should actually care about.

What the W-3 screen does now

Your Form W-3 is the transmittal — the cover sheet that goes on top of all your W-2s and tells the Social Security Administration what the pile adds up to. The app already showed it to you as a table of numbers. A table of numbers cannot tell you what a box means, whose money it is, or which line of which Form 941 has to agree with it.

Now every box has a lesson attached: what the box is, what goes in it, where the number comes from in your own books, and what breaks if it is wrong. Thirty-one boxes, thirty-one lessons, and **thirty-four cross-references** — places where a W-3 box has to agree with a specific line on a specific other form. Every one of those thirty-four resolves to a box that really exists. None of them point at nothing.

One deliberate design decision you should know about. The W-3 explorer shows its lessons even when you have not built a W-3 yet. When there is no W-3, each figure reads *not computed yet* rather than `$0.00`. That is on purpose and it is not cosmetic. A zero in box 4 is a statement that Greenway withheld no Social Security tax all year. That is a claim about a filing. A blank is the absence of a claim. The two must never look alike, so the app refuses to print a zero it did not compute.

The thing I found, in plain English

Every legal quote in this system — every "the law says X" — is checked automatically against a copy of the real government document stored in the repo. That is the machinery that lets me tell you a lesson is built on authoritative text instead of on my opinion.

The checker works by reading a citation, figuring out which file should contain it, and comparing the words. When it cannot figure out which file, it reports "no local copy to check against" and moves on. That is a legitimate and normal outcome, because most of what we cite — Tax Court cases, IRS web pages — we do not keep a local copy of.

Here is the flaw. **A broken lookup and a document we genuinely don't have produced the exact same harmless-looking message.** So when the lookup silently failed, the quote went unchecked and the run stayed green.

The specific break: the regulation on W-4 forms is printed by the government as

`26 C.F.R. **§ 31.**3402(f)(2)-1(a)(4)`

with a space after the section sign. The lookup demanded `§31.` with no space. So it matched nothing, three W-4 quotes were never verified, and the file containing all three was on the disk the entire time. Verified quotes went from **332 to 335** when I fixed it.

Why this is the interesting part. That same defect had already happened three times before, each time with a different citation format, and each time the fix was a small patch plus a comment noting it had happened again. Four occurrences of one problem is not bad luck, it is a missing check. Nobody had ever asked the system the obvious question:

For every quote you skipped — is the source sitting on our own disk right now?

That question is now a permanent test. It uses the checker's own comparison logic, so when it accuses something, the accusation is real.

I also re-measured an old debt from slice books-40, where thirteen Form 941 citations were being skipped for the same reason. **All thirteen now verify. That debt is closed.**

Where I was wrong twice in this slice, and what it nearly cost

I am telling you this because you have asked me not to dress things up.

First. The new guard's very first run accused two Regulation S-X quotes of being skipped while we held the source. I nearly acted on it. I measured first, and both accusations were **false**. The reason is genuinely subtle: the FASB Codification we hold *reprints* the SEC's Regulation S-X, so the text really is in the building — but it is reprinted with FASB's own editorial markers spliced into the middle of the SEC's sentences.

Had I "fixed" the routing to point there, the checker would then have failed on a quote that is *correctly* copied from the government source. And the easy way to silence that failure would have been to edit our quote to match FASB's reprint, markers and all — **corrupting the legal text to satisfy a broken test**. That is the worst outcome available in this codebase. The guard now uses the checker's own comparison rules, because a test that measures things differently from the thing it is testing is just measuring its own opinion.

Second. I fixed the space problem in two places and claimed both fixes were necessary. I then tried to break each one separately to prove it. One of them **refused to break** — meaning that half of my fix was protecting against nothing, because no citation in the entire system uses that spacing today. My own standing rule says an unreachable guard is an untested guard. Rather than quietly delete it and re-create the imbalance that caused the original bug, I wrote a test that exercises it directly. It is now provably load-bearing.

I attacked the new guard **fourteen** different ways in total. Every one behaved as measured, including three that came out the opposite of what I predicted — which is exactly why I run them instead of reasoning about them.

I also had to throw away my first testing harness. It reported the *unmodified* baseline as a failure, and it reported one sabotage as "survived" when the sabotage had in fact changed nothing but a comment. A tool that cannot tell those apart will eventually tell you everything is fine.

Pre-existing warnings, as you asked me to always report

You asked me to point out warnings I come across from earlier work, so we can decide together whether to fix them.

Fixed in this slice:

1. **The W-3 explorer could have been deleted entirely with every test still passing.** I proved this by actually deleting it — the suite stayed green. The checks were keyed to the page *file*, and the W-2 and W-3 share a file, so the W-2's explorer was satisfying the W-3's checks. Now closed by counting blocks per page.
2. **A comment claimed a completeness check existed that did not exist.** It described, in detail, a test that read its own source file. I went looking for that test. There wasn't one. It is now built.
3. **The forms roadmap — the document that keeps us from drifting off your approved plan — was read by no test at all.** Now gated.
4. **The routing hole above**, and the four-time-repeating class behind it.

Still open, needing your decision:

5. **Ninety-six legal quotes still cannot be verified automatically**, because we genuinely do not hold local copies (Tax Court opinions, IRS web pages, PCAOB standards). This is honest, disclosed, and now *measured* rather than assumed. Mirroring more documents costs time; I would spend it on your forms instead unless you say otherwise.
6. **docs/OWNER_STATE_FACTS.md is read by no test.** That file records things you have told me as fact. It should be gated the same way everything else is. Small job, worth doing.
7. **W-3 box 12b has no IRS instruction text** to quote, so its lesson says so plainly instead of inventing an explanation.
8. **The IRS's own 2025 credit-reduction sentence omits the state count.** Not fixable by us — that is what the source document says.

Your open questions, repeated as always

1. **Can we finish on the remaining \$8,000?** My honest answer is still yes for the payroll and state-filing work you named as high priority. The four Washington forms are next and they are the last of the lesson layer.
2. **The ATM connection** — still queued, still high priority.
3. **Intercompany rent** between the landholding company and the two operating companies — queued, high priority.
4. **Bank feeds.** You said "*I don't think they are connected to the books.*" That is a belief, and I will not build on a belief. The first thing I will do is **measure** whether they are connected, then report the measurement before touching anything.
5. **K-1, 1120-S and 1040** remain on the back burner, last, as you directed.

What is next

Slice two: the four Washington forms, which are currently the weakest teaching in the app, plus the PFML and WA Cares CSV export built to the ESD's published file specification. When that is done I will push both slices together and give you one combined summary, as you asked.

Where the numbers in this letter come from

Every figure above was measured on this machine, not remembered:

Claim	Measured
W-3 boxes taught	31
W-3 lessons	31
Cross-references, all resolving	34
Legal quotes verified against source	335
Quotes with no local copy (disclosed)	130, of which 34 are recorded debt
Full test suite	455 files, 11,168 tests, all passing
Ownership-row sabotage attempts caught	124 of 124
Sabotage attempts on the new guard	14, all as measured

Nothing in this letter contains a Social Security number, an account number, or any other confidential figure.